

## REUS AP, Spain

WMO: 081750

Lat: 41.1497N

Lon: 1.1789E

Elev: 76

StdP: 100.42

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.8       | 0.9 | -10.9                         | 1.5 | 6.3  | -7.8 | 2.0 | 4.9  | 14.5                  | 12.2 | 13.4 | 12.6 | 1.9                   | 20   | 0.398 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.4                    | 32.4            | 22.1 | 31.2 | 22.1 | 30.2 | 21.9 | 24.8                | 29.2 | 24.2 | 28.7 | 23.6 | 28.1 | 4.5                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.6                            | 18.6 | 27.4 | 22.8 | 17.7 | 27.0 | 22.1 | 16.9 | 26.7 | 75.8          | 29.2 | 73.0 | 28.9 | 70.7 | 28.1 | 27.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 10.8              | 9.1  | 7.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.7   | 8.9   | 9.7   | 11.9  | 14.5  | 18.0  | 22.3  | 25.4  | 25.5  | 22.1  | 18.4  | 12.8  | 10.1 |
|                                                                                  | DBStd       | 6.42   | 3.15  | 2.99  | 2.40  | 2.15  | 2.41  | 2.49  | 1.79  | 1.84  | 2.41  | 2.76  | 3.30  | 2.78 |
|                                                                                  | HDD10.0     | 147    | 57    | 36    | 9     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 12    | 32   |
|                                                                                  | HDD18.3     | 1344   | 293   | 241   | 200   | 117   | 36    | 2     | 0     | 0     | 3     | 32    | 167   | 254  |
|                                                                                  | CDD10.0     | 2584   | 22    | 28    | 67    | 135   | 247   | 368   | 478   | 482   | 363   | 262   | 96    | 36   |
|                                                                                  | CDD18.3     | 740    | 0     | 0     | 0     | 1     | 24    | 119   | 219   | 223   | 116   | 36    | 1     | 0    |
|                                                                                  | CDH23.3     | 5888   | 0     | 0     | 2     | 11    | 136   | 901   | 1938  | 1987  | 742   | 171   | 1     | 0    |
|                                                                                  | CDH26.7     | 1801   | 0     | 0     | 0     | 0     | 11    | 249   | 683   | 690   | 151   | 17    | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.0    | 3.4   | 3.4   | 3.3   | 3.0   | 3.0   | 2.7   | 2.8   | 2.6   | 2.7   | 2.6   | 3.2   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 562    | 35    | 31    | 39    | 42    | 53    | 43    | 21    | 51    | 66    | 86    | 55    | 40   |
|                                                                                  | PrecMax     | 959    | 144   | 100   | 186   | 217   | 124   | 190   | 92    | 270   | 266   | 278   | 215   | 148  |
|                                                                                  | PrecMin     | 338    | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 165    | 31    | 24    | 36    | 44    | 32    | 45    | 26    | 54    | 60    | 70    | 52    | 39   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 20.7  | 21.3  | 22.4  | 24.8  | 27.9  | 33.5  | 33.4  | 35.1  | 31.2  | 28.9  | 22.6  | 19.2 |
|                                                                                  |             | MCWB   | 13.6  | 13.3  | 13.3  | 15.7  | 17.5  | 21.7  | 21.7  | 22.4  | 21.5  | 17.2  | 16.3  | 13.6 |
|                                                                                  | 2%          | DB     | 18.2  | 18.8  | 20.6  | 22.6  | 26.4  | 30.9  | 32.1  | 32.5  | 29.5  | 26.2  | 21.1  | 18.0 |
|                                                                                  |             | MCWB   | 11.8  | 11.8  | 12.6  | 14.7  | 16.9  | 20.5  | 22.2  | 22.5  | 21.1  | 18.4  | 15.9  | 12.6 |
|                                                                                  | 5%          | DB     | 16.7  | 17.0  | 19.1  | 21.1  | 24.9  | 29.4  | 31.1  | 31.1  | 28.3  | 25.1  | 19.9  | 16.8 |
|                                                                                  |             | MCWB   | 11.1  | 10.7  | 12.0  | 14.4  | 16.9  | 20.5  | 22.5  | 22.2  | 20.8  | 18.7  | 15.1  | 11.8 |
|                                                                                  | 10%         | DB     | 14.9  | 15.5  | 17.7  | 19.8  | 23.3  | 27.9  | 30.1  | 30.1  | 27.1  | 23.9  | 18.3  | 15.5 |
|                                                                                  |             | MCWB   | 10.0  | 10.1  | 11.4  | 13.7  | 16.2  | 19.8  | 22.3  | 22.2  | 20.2  | 18.3  | 13.7  | 11.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.0  | 13.9  | 15.1  | 16.7  | 19.5  | 23.8  | 25.5  | 26.0  | 24.2  | 21.6  | 18.5  | 14.7 |
|                                                                                  |             | MCDB   | 19.3  | 19.0  | 19.9  | 21.8  | 24.2  | 29.9  | 29.7  | 30.1  | 28.5  | 25.0  | 20.7  | 17.4 |
|                                                                                  | 2%          | WB     | 12.8  | 12.8  | 13.8  | 15.9  | 18.7  | 22.8  | 24.7  | 25.2  | 23.3  | 20.7  | 17.2  | 13.5 |
|                                                                                  |             | MCDB   | 16.9  | 17.0  | 18.5  | 20.7  | 23.6  | 28.7  | 29.3  | 29.2  | 27.3  | 24.0  | 19.8  | 16.5 |
|                                                                                  | 5%          | WB     | 11.7  | 11.8  | 13.0  | 15.3  | 18.0  | 21.8  | 24.1  | 24.4  | 22.5  | 20.0  | 16.0  | 12.8 |
|                                                                                  |             | MCDB   | 15.6  | 15.7  | 17.2  | 19.5  | 22.4  | 26.9  | 28.9  | 28.6  | 26.4  | 23.0  | 18.7  | 15.5 |
|                                                                                  | 10%         | WB     | 10.7  | 10.9  | 12.2  | 14.6  | 17.3  | 20.9  | 23.5  | 23.8  | 21.8  | 19.3  | 14.7  | 11.8 |
|                                                                                  |             | MCDB   | 14.2  | 14.1  | 16.2  | 18.3  | 21.7  | 26.0  | 28.2  | 28.1  | 25.6  | 22.1  | 17.3  | 14.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.2  | 10.4  | 10.6  | 9.7   | 10.3  | 10.0  | 9.2   | 9.4   | 9.8   | 9.9   | 10.0  | 10.0 |
|                                                                                  |             | MCDBR  | 12.0  | 12.9  | 13.5  | 12.2  | 12.0  | 11.6  | 10.7  | 10.8  | 10.9  | 11.3  | 10.9  | 11.3 |
|                                                                                  |             | MCWBR  | 7.5   | 7.6   | 7.6   | 6.2   | 5.6   | 5.3   | 5.1   | 5.5   | 5.4   | 6.0   | 6.5   | 7.2  |
|                                                                                  | 5% WB       | MCDBR  | 10.7  | 11.5  | 11.4  | 10.3  | 9.7   | 10.3  | 9.2   | 8.9   | 9.5   | 9.2   | 8.7   | 10.5 |
|                                                                                  |             | MCWBR  | 7.1   | 7.3   | 7.1   | 5.8   | 5.0   | 5.0   | 4.8   | 4.8   | 5.0   | 5.5   | 5.9   | 7.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.313  | 0.333 | 0.378 | 0.397 | 0.418 | 0.433 | 0.437 | 0.437 | 0.414 | 0.385 | 0.339 | 0.314 |      |
|                                                                                  | taud        | 2.518  | 2.464 | 2.351 | 2.316 | 2.295 | 2.292 | 2.298 | 2.298 | 2.356 | 2.427 | 2.498 | 2.519 |      |
|                                                                                  | Ebn at Noon | 847    | 878   | 869   | 875   | 861   | 846   | 838   | 827   | 822   | 806   | 805   | 813   |      |
|                                                                                  | Edh at Noon | 73     | 89    | 112   | 124   | 130   | 130   | 128   | 125   | 111   | 92    | 75    | 68    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.10   | 3.04  | 4.38  | 5.48  | 6.50  | 7.26  | 7.23  | 6.21  | 4.79  | 3.39  | 2.29  | 1.82  |      |
|                                                                                  | RadStd      | 0.17   | 0.34  | 0.34  | 0.37  | 0.43  | 0.23  | 0.23  | 0.23  | 0.23  | 0.26  | 0.22  | 0.14  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +107    | +84     |

Nomenclature: See separate page